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Automata

Documentation For Final Project

I started this project wanting to do something fun and have a system that would either randomly or, if the teacher wanted, specifically pick out a student, throw them into a game of the teacher's choice against the teacher. And if the student won, they could go free, but if they didn't, then they would have to face some random punishment that would be selected. For this project, I realized that I wanted to create 4 separate files, each with its own state, in order to make it more manageable to handle each aspect of the project. I started by creating my students' text file, where I just copied each name from Canvas and put it into a file. I then created a student manager file that would go into the text file and either check to see if the name the teacher entered was there or randomly select a name if that's what the teacher wanted instead. I then created the game file, which was the largest and most difficult one to deal with. I first had to set up a play game variable that checked and saw what game the user wanted to pick. It also had to make sure the user actually selected an option. I then created three functions, one for each game: rock, paper, scissors, guess the number, and Tic Tac Toe. The first two were easy, as I would just check the teacher's answer against the student's. However, Tic Tac Toe required an entire board to be created, requiring other functions in order for it to work and look good as well. Then I created the exile file, which has a string of punishments and a string of rewards just in case my name gets selected. If the student loses the game, they get randomly punished, but if I lose, then I get randomly rewarded. The final main file is then just as simple as I ask the teacher if he would like to choose randomly or pick out a student and then go through the student manager file, then the game file, then the exile file. The program then keeps repeating until the teacher decides to enter q to quit it.